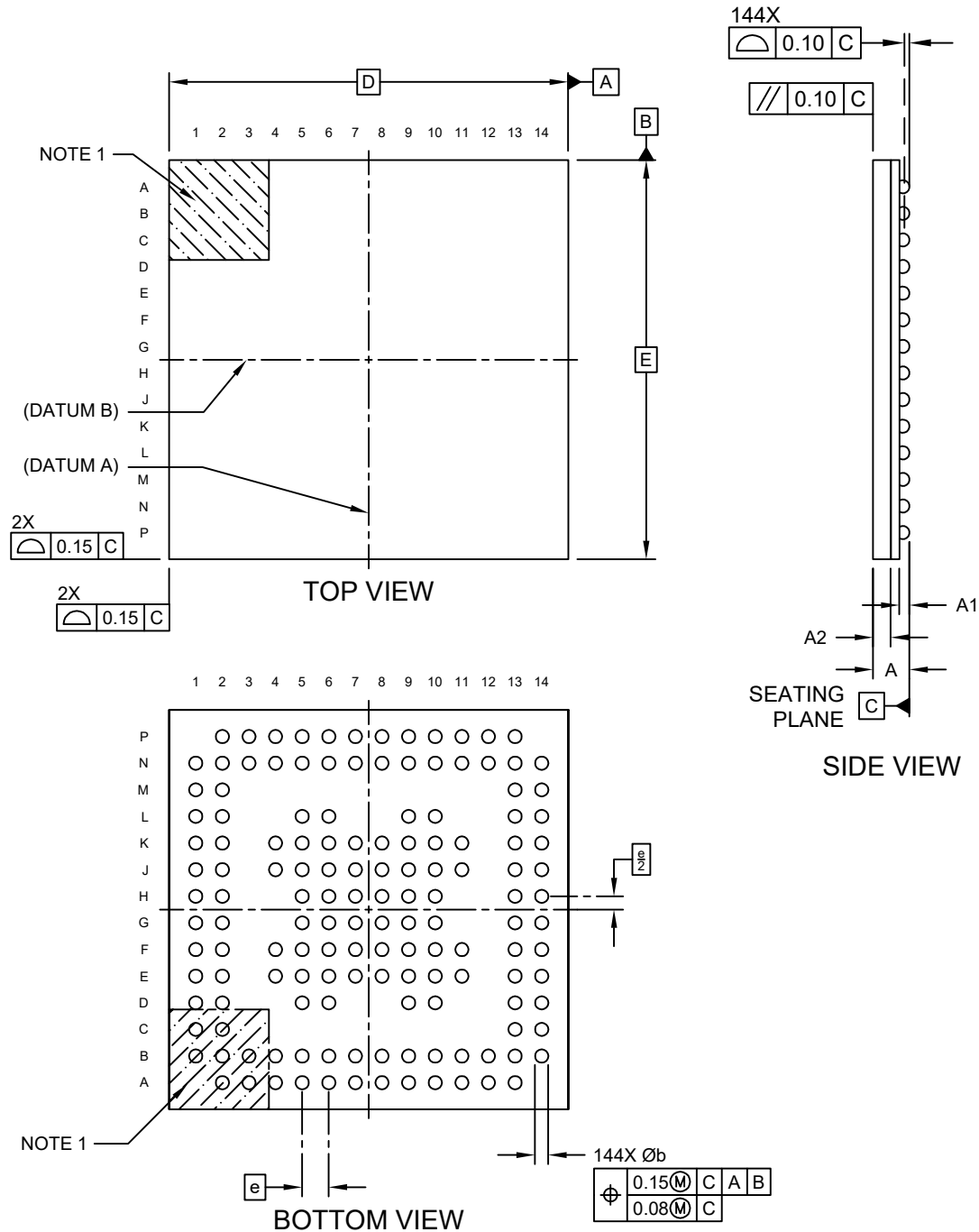


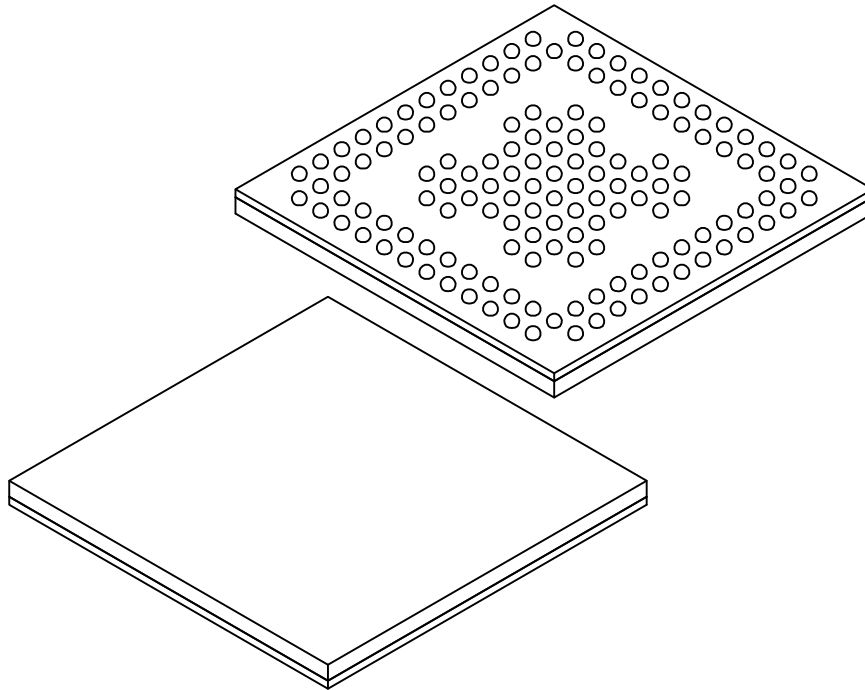
144-Ball Thin Fine-Pitch Ball Grid Array Package (LUX) - 12×12×1.19 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



144-Ball Thin Fine-Pitch Ball Grid Array Package (LUX) - 12×12×1.19 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



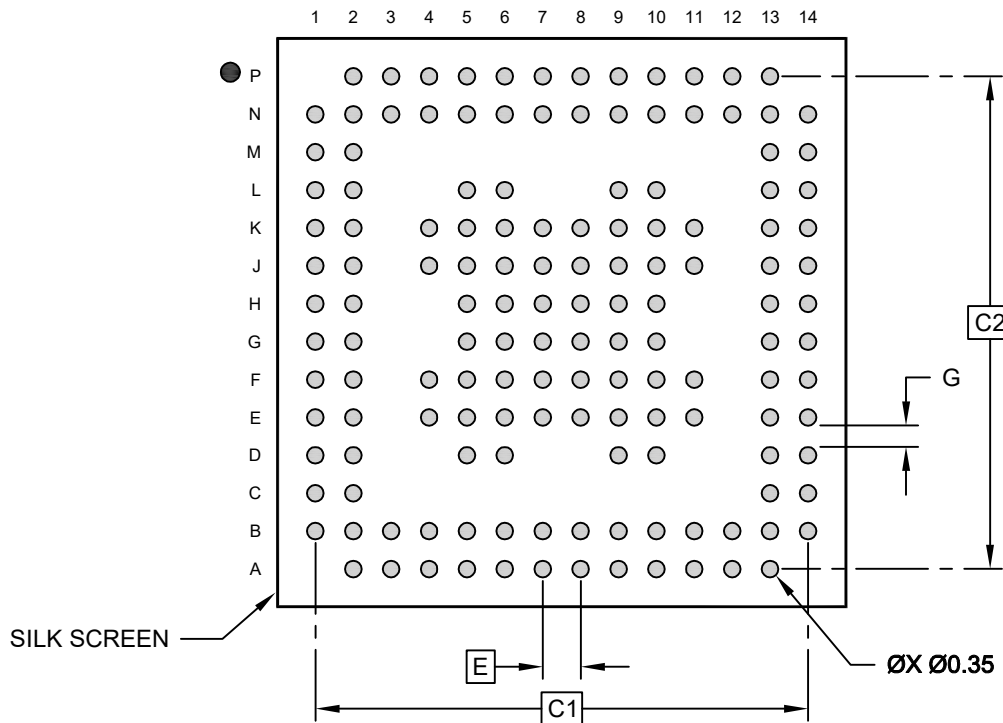
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	144		
Pitch	e	0.80 BSC		
Overall Height	A	–	–	1.19
Ball Height	A1	0.21	0.30	–
Mold Thickness	A2	0.48	0.53	0.58
Overall Length	D	12.00 BSC		
Overall Width	E	12.00 BSC		
Ball Diameter	b	0.35	0.40	0.45

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

144-Ball Thin Fine-Pitch Ball Grid Array Package (LUX) - 12×12×1.19 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1	10.40 BSC		
Contact Pad Spacing	C2	10.40 BSC		
Contact Pad Width (X144)	X			0.35
Contact Pad to Center Pad	G	0.45		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2511 Rev A